



GIA - IAA

Introducció a

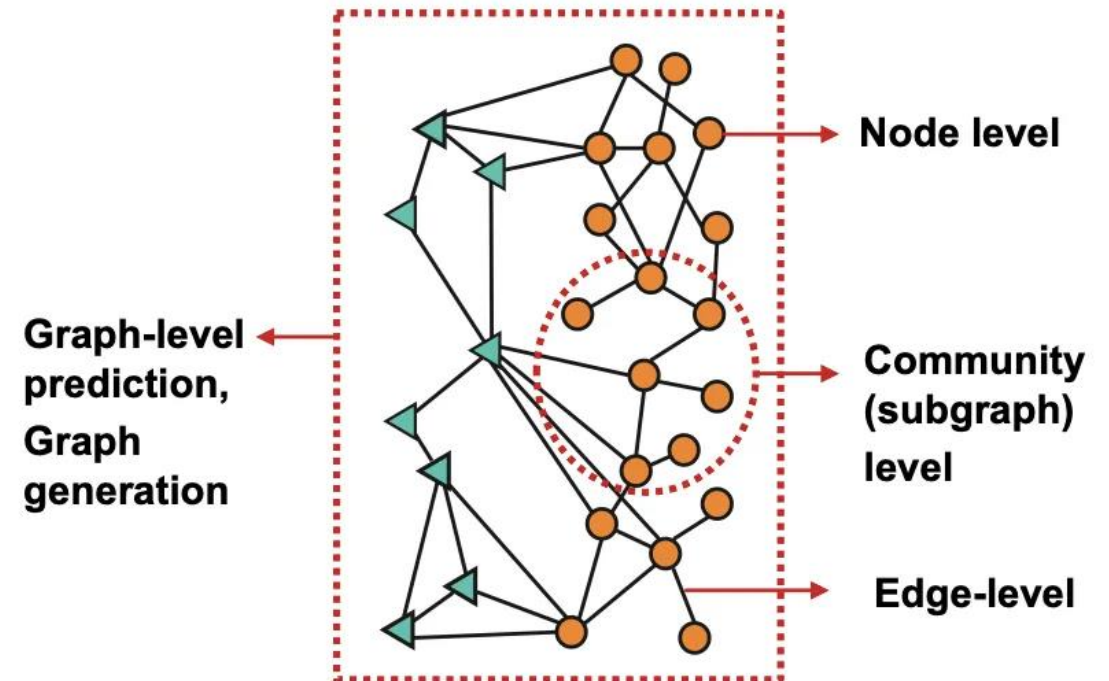
l'Aprenentatge Automàtic

10. Lab: Graph Learning

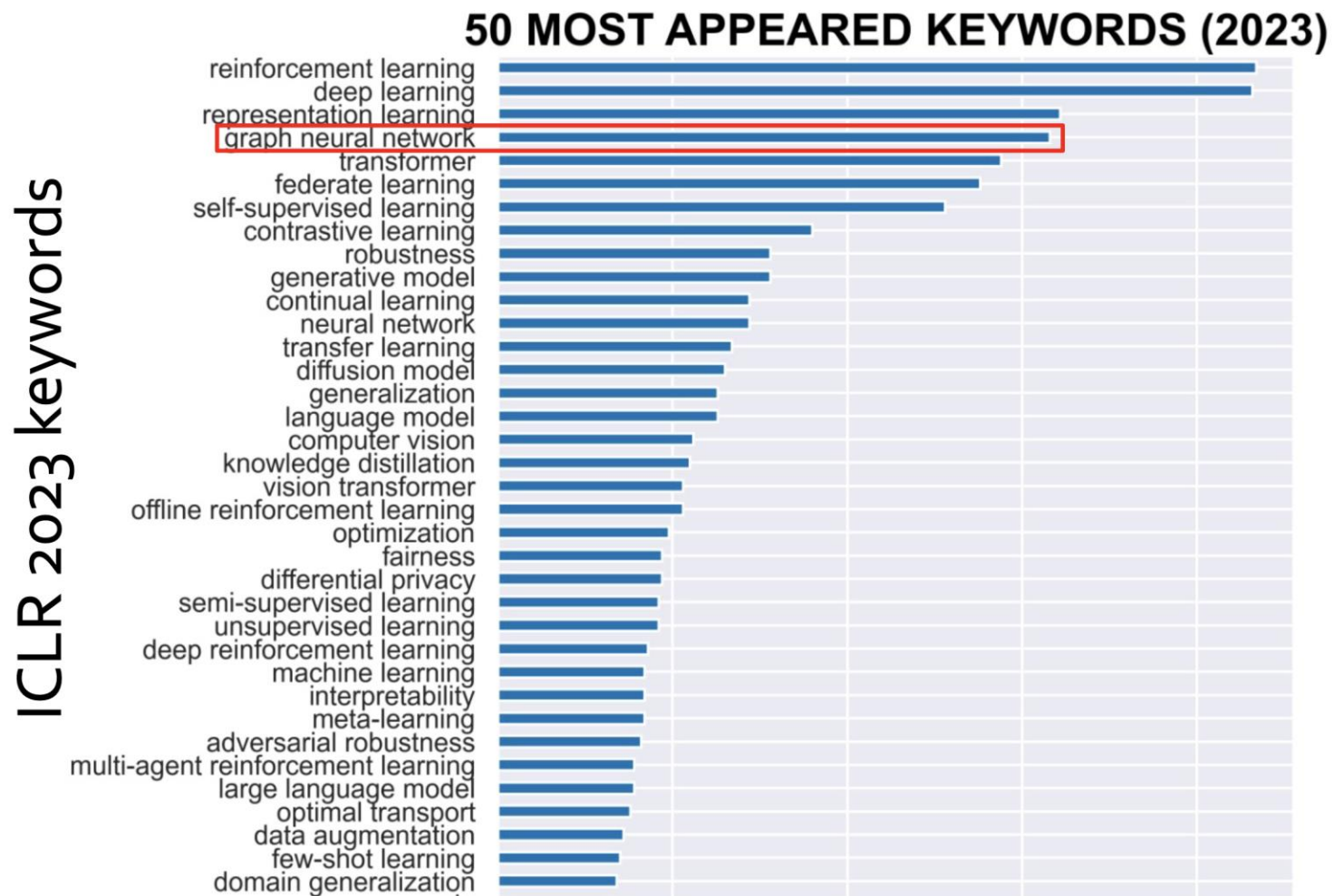
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Why graphs?

- ❖ Graphs are a general language for describing and analysing entities with relations/interactions.

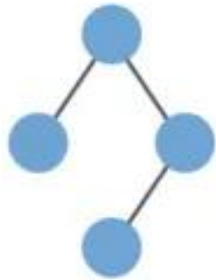


Why graphs?

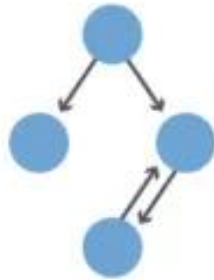


Kind of graphs

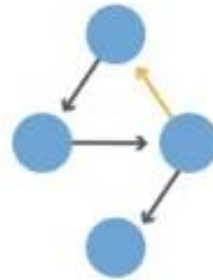
Undirected



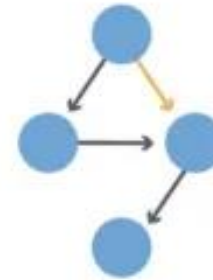
Directed



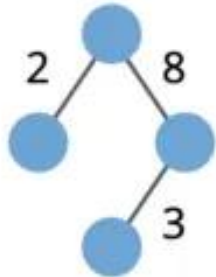
Cyclic



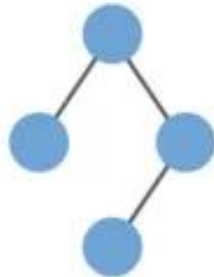
Acyclic



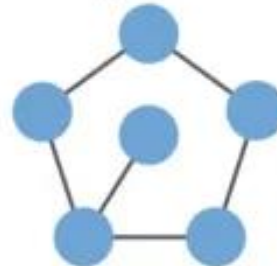
Weighted



Unweighted



Sparse



Dense



Networkx library



```
1 # Crear un graf no dirigit
2 G = nx.Graph()
3 print(G.is_directed())
4
5 # Crear un graf dirigit H
6 H = nx.DiGraph()
7 print(H.is_directed())
8
9 # Afegir atributs a nivell de graf
10 G.graph["Name"] = "I am graph!"
11 print(G.graph)
```

[277] ✓ 0.0s

```
... False
True
{'Name': 'I am graph!'}
```

Contact

[Mailing list](#)
[Issue tracker](#)
[Source](#)

Releases

Stable (notes)

3.2.1 — October 2023
[Documentation](#)

Latest (notes)

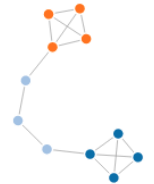
3.3 development
[Documentation](#)

Archive



NetworkX
Network Analysis in Python

NetworkX is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

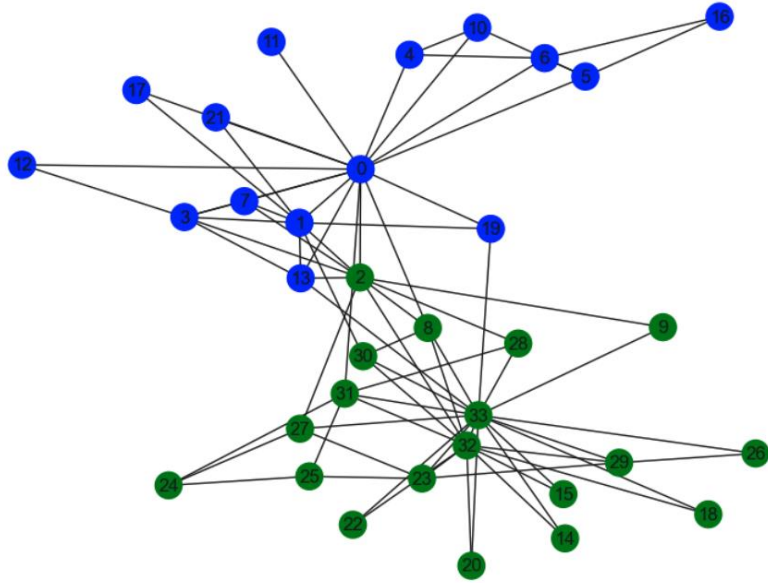


Software for complex networks

- Data structures for graphs, digraphs, and multigraphs
- Many standard graph algorithms
- Network structure and analysis measures
- Generators for classic graphs, random graphs, and synthetic networks
- Nodes can be "anything" (e.g., text, images, XML records)
- Edges can hold arbitrary data (e.g., weights, time-series)
- Open source [3-clause BSD license](#)
- Well tested with over 90% code coverage
- Additional benefits from Python include fast prototyping, easy to teach, and multi-platform

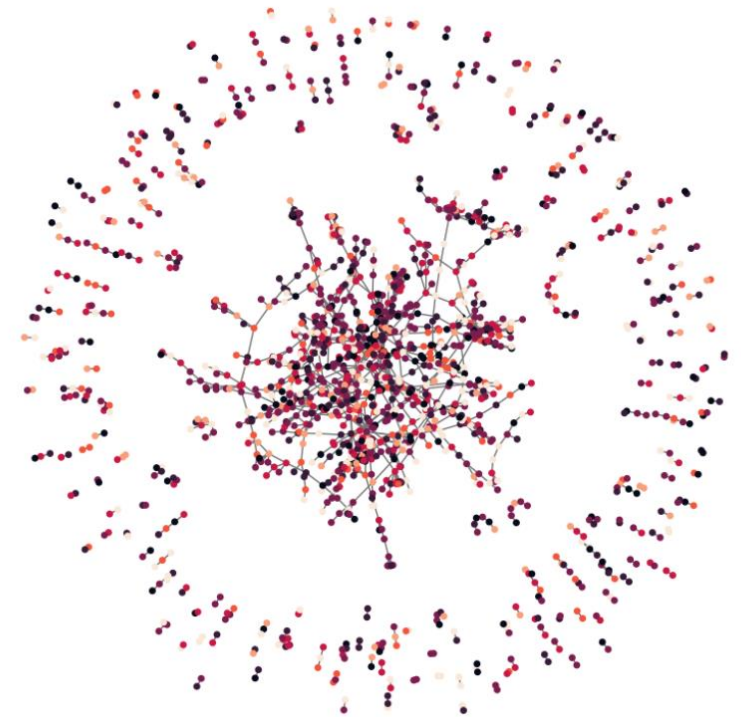
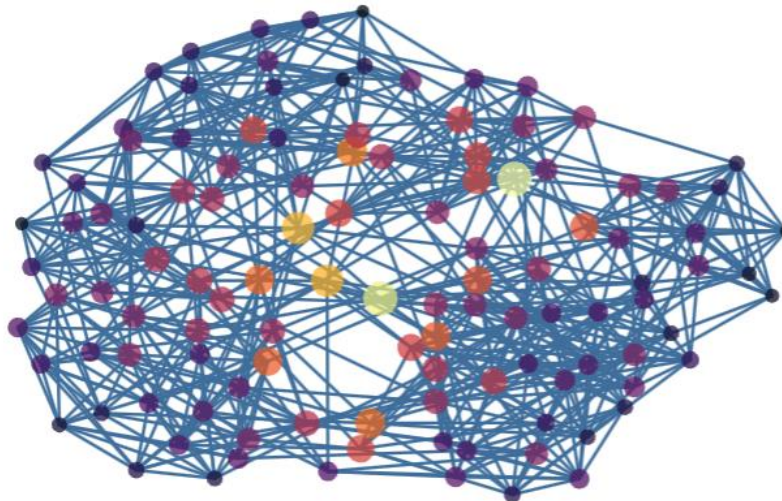
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Some task examples



Identifying communities, e.g, based on the relations/interactions in a social network.

Identifying centrality of nodes in a network, e.g. football teams.

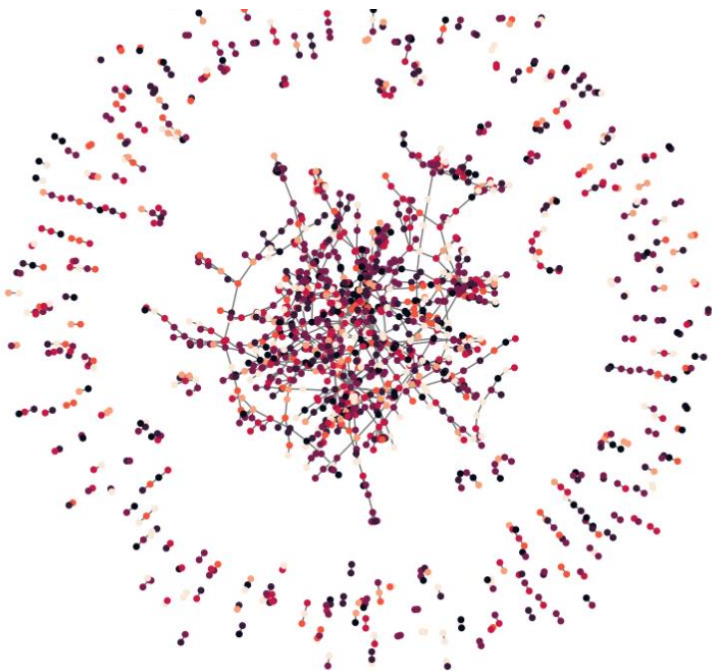


Predicting node labels or link between nodes, e.g., in a paper citation network.

Some task examples

Playtime

PLAYTIME: Repeat the node prediction task, seen in the previous section. For doing so, modify the Node2Vec embedding method to create an embedding that includes both the paper content and the network topology (citations) for the whole cora dataset. Idea: Combine the previous computed embeddings and the features based on the paper content. Including the network relationship improve the performance in the node classification? How the dimension of the embedding affects the node prediction performance?



Predicting node labels using node embeddings computed from the network topology augmented with the node attributes.